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Project Title:	Eutectic Solidification in a Ternary Alloy		
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Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
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	Note this is a follow-on project for Dr. Ralph Napolitano's project,"Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy," which ended in 2014. Solidification processing of metallic alloys can produce a wide array of growth morphologies, which can be manipulated to access new structures, materials, and functionalities. The array of accessible structures is particularly wide in ternary and other higher-order systems, but the challenge of controlling the many variables that influence growth structures is immense and requires a better understanding of the interplay between the mechanisms of pattern formation and the influence of applied forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration,		

Task Description:	faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal field. The two principle objectives to be addressed are (i) to establish an analytical framework for 2D and 3D quantitative description of multiphase microstructures, and (ii) to employ the newly developed quantitative framework to investigate the dynamics of pattern formation in ternary eutectic alloys. The work will focus primarily on the invariant three-phase eutectic structure. Ground-based samples will be processed via a vertical Bridgman-style directional solidification furnace. Electron backscatter diffraction will be a major characterization tool to measure crystallographic orientation, texture, and misorientation relationships. Analysis over the length of the sample will provide insight into initial transient behavior, grain competition, and steady state growth. The relationship between geometric and crystallographic parameters, such as curvatures, orientations, and periodic ordering, will be used to quantify the morphological trajectory of the eutectic and identify fundamental principles of pattern formation. In order to validate the experimental correlations and analytical models developed from the findings of the ground-based work, carefully selected flight-based directional solidification experiments will be conducted using the European Space Agency (ESA) Materials Science Lab (MSL) and the Low Gradient Furnace (LGF). The project operates in collaboration with the European Space Agency's project titled "Solidification along a Eutectic Path in Ternary Alloys" (SETA), which has already accomplished important work using the Al-Ag-Cu alloy system. Significant findings of the SETA project to date have been in the areas of pattern development and the role of fault structure for both univariant and invariant eutectic compositions, the role of crystal orientation and "locked" grains, and development of nonplanar growth modes. Facilitating open scientific interaction between the Principal Investigator (PI) and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. The planning and performance of flight-based solidification experiments is being done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled, multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is far-reaching, given the broad range of multiphase structures that can be fabricated through solidification processing, and is not limited to metals and alloys. Indeed, the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
	Specific areas of focus during Year 5 have been (i) the effects of solid-state diffusion on pattern evolution in Al-Ag-Cu three phase eutectic, (ii) the relationship between crystallographic orientation relationship and eutectic grain competition, (iii) continuing evaluation of data obtained by tomography, with particular emphasis on grain competition, (iv) novel methods of characterizing complex three-phase structures, and (v) analysis of the effect of thermal gradient on complex eutectic growth, using the Al-Cu-Mg system for comparison. The work made extensive use of the Bridgman furnace built in Year 1 and the automated image segmentation and analysis tools developed during Years 2-4. 1.1 SOLID STATE CHANGES IN AL-AG-CU TERNARY EUTECTIC One challenge in studying the ternary eutectic system Al-Ag-Cu is a relative large solubility shift of the chemical element Ag in the Al phase after solidification, which influences both the phase fractions and composition. One way to observe these effects is to quench the sample and observe the microstructure at different distances behind the quenched interface, which has been done previously by members of the SETA project team. While this allows observation of microstructure evolution within a particular grain, it does not permit a specific region of microstructure to be observed at various stages. It is necessary to clearly understand the effect of these solid-state changes in order to separate them from the original solidification behavior. This is particularly important for the evaluation of flight samples, since those samples were processed without quenching, and only microstructure influenced by solid state diffusion is therefore available for analysis. To directly study the effects of solid state diffusion, directionally solidified samples of ternary eutectic Al-Ag2Al-Al2Cu were quenched. Then, the samples were systematically annealed just below the eutectic temperature and the microstructure quantitatively analyzed. This mimics the process that occurs naturally during cooling in the directional solidification furnace. We found a good comparison between the patterns after annealing and those seem in unquenched directionally solidified samples. Different patterns exhibited different reactions during annealing, which results in a different high-temperature stability. For the chain-like structure, the pattern is preserved even after annealing for 32 h at more than 96% of the melting temperature, which underlines the capability of eutectics for high temperature applications. The interfaces become straighter and more aligned, which demonstrates the influence of the crystallographic anisotropy. For the cross structure the pattern is not preserved; instead the rods merge together and the interfaces become rounder. It can also be seen that phase fraction adjusts much more quickly than shape factor, which could be an important factor in deciding which parameter(s) to use to characterize and compare different microstructures. 1.2 ROLE OF ORIENTATION RELATIONSHIPS IN EUTECTIC GRAIN COMPETITION From previous work, it has become clear that eutectic grains with different patterns and different crystallographic orientation relationships (ORs) can form in directionally solidified invariant Al-Cu-Ag. It has also been observed that the eutectic patterns are more varied and irregular at the beginning of solidification compared to the microstructures observed after longer solidification distances. Therefore, we undertook to study the way that grain competition during growth affects the evolution of the microstructure, and particularly what role ORs play in that competition. A sample of the ternary eutectic composition was directionally solidified at a velocity of 0.625 um/s and thermal gradient of 15 K/mm. The sample was cut to reveal transverse cross sections at ten solidification heights. Each surface was polished and a high resolution montage of the full circular cross section was created. The grain boundaries were manually identified, based on differences in patterns and pattern alignment between grains, and drawn onto the images by hand.

Task Progress:	The number of grains per section was calculated and the types of patterns present were quantified. As expected, the number of grains dropped rapidly, as those grains containing the most highly aligned chain-like pattern expended to cover the entire surface of the sample. Electron backscatter diffraction (EBSD) was carried out on the surfaces at 7 and 47 mm to determine the orientation relationships of the grains. Analysis of these results is ongoing. 1.3 TOMOGRAPHY FOR 3D ANALYSIS OF TERNARY EUTECTIC GROWTH During Year 4, extensive tomographic data was collected at the Advanced Photon Source at Argonne National Lab from samples processed at UAB (University of Alabama at Birmingham). Two regions (diameter = 0.5 mm, length = 6-7 mm) were machined from a sample processed at various conditions and subjected to tomography. Analysis began during Year 5 with investigation of grain competition and the mechanisms for lateral grain expansion. The 3D data makes it very clear where the grain boundaries are, including identifying boundaries that may not be noticeable from a single cross section. The current work is focusing on the role of tilt on grain competition during solidification, using samples solidified at the lowest velocity (0.37 μm/s) to create the largest microstructure for easiest viewing. Previous analysis identified that the ternary eutectic structure often does not grow with the interphase boundaries perfectly aligned to the direction of heat extraction, but this is difficult to quantify without three dimensional data. The tomography confirmed first that this tilt is real, rather than an artifact of cutting or mounting the sample at an angle, and that the tilt varies from grain to grain. In the chain-like pattern most common to Al-Ag-Cu, especially at low growth velocities, it was discovered that this misalignment can be characterized by two different angles. Tilt of the boundary between the intermetallic phases causes the chain “links” (Al2Cu and Ag2Al) to appear to run forward or backwards along the length of the chain as one scrolls through the 3D stack of images. Tilt of the boundary between the intermetallic chain and the lamellar aluminum causes the chains themselves to drift left or right. These two tilt angles are constant within a eutectic grain, and will be measured by tracking the position of individual phase regions within different grains across different tomography slices. The tilt angles can be verified by measurements taken on images parallel to the growth direction. The hypothesis is that grains which have more tilt will be overgrown and disappear from the sample more quickly. 1.4 NOVEL TECHNIQUES FOR CHARACTERIZING COMPLEX MICROSTRUCTURES Finding ways to characterize the complex, multi-phase microstructures found in ternary eutectics is a constant challenge. Unlike regular binary eutectics, the structures cannot be described by a single scalar value (lamellar spacing). We are therefore continually looking for new methods of manipulating and quantifying these microstructures in order to better understand the underlying physics that controls their behavior. To that end, the postdoc employed on this project spent time during Year 5 creating new MatLab code to analyze the micrographs in new ways. This builds heavily on the image segmentation and quantification routines developed in MatLab specifically for three phase microstructures that were written by group members in previous years. An example of one of these new characterization methods and how it helps to better understand the behavior of the material, is a graph measuring the angle of the boundary between the Al2Cu and Ag2Al phases with respect to a horizontal line. The graphs were used in Steinmetz et al., J Crystal Growth 2019, to show how the interfaces in a grain that is initially well aligned become more aligned to one particular angle with increased annealing time. For this structure, nearly all of the interfaces have an alignment between 45 and 95 degrees, whereas a more poorly aligned pattern begins with a much more uniform distribution of curvature, with peaks at 5 and 95 degrees. In this case, both peaks shrink with annealing time, indicating that the interface boundaries favored during growth are not the same as those favored during annealing. A small peak around 35 degrees begins to appear at longer annealing times, which suggests that a crystallographically favorable low energy boundary may lie along that position. Further examination with EBSD is necessary to support that conclusion. Another new method uses renders the two intermetallic phases of the microstructure as a series of circles (Ag2Al) and squares (Al2Cu). In a perfect version of this pattern, each region would have two nearest neighbors of the opposite phase. Phase regions with three nearest neighbors are given in red, while phases with only one neighbor, that is, forming the end of a chain, are drawn in blue. This allows for easy visual identification of the different types of faults in the pattern, and how they change with position or time. Additionally, micrographs were colored to highlight changes in shape factor, alignment and boundary curvature. 1.5 EFFECT OF THERMAL GRADIENT ON COMPLEX EUTECTICS Although thermal gradient has a significant effect on dendritic growth, it is generally considered not to influence eutectic solidification, and is not included in the classic Jackson-Hunt equation for eutectic growth. However, previous results from various members of the SETA team using different furnace setups have shown that large differences in thermal gradient do influence the structure of ternary eutectic observed in Al-Ag-Cu. In order to investigate these effects, as well as undertake comparison of another eutectic system, experiments were planned using the Al-Cu-Mg system. This alloy has been studied much less than the Al-Ag-Cu system, but provides a good comparison, with two of the three phases being the same (solid solution Al and Al2Cu) and the third being an intermetallic with a very different crystal structure (Al2CuMg, orthorhombic compared to Ag2Al, hexagonal). A number of experiments have been completed, with the microstructures being imaged and the images quantitatively analyzed. Analysis of those results is currently underway.

Bibliography Type:	Description: (Last Updated: 06/05/2023)
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